

From Solid-Earth Tides to $\Delta v/v$ at SAFOD

A Detailed Theory and Modeling Guide for the 2026 AWD Experiment

Working notes for group-meeting preparation

August 2026

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How equation provenance is labeled in this guide

This revision distinguishes six different kinds of statements that were too easy to mix together in the original notebook and guide:

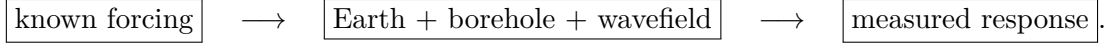
- **[SOURCE – Métivier & Conrad (2008)]**: an equation, framework, or numerical value explicitly given in that paper, or a direct component-wise rewriting of it.
- **[SOURCE – Niu et al. (2008)]**: a measured or calculated numerical value reported by Niu et al. for their SAFOD experiment.
- **[SOURCE – Hole et al. (2001)]**: fault-geometry information used to motivate a Parkfield geometry scenario.
- **[STANDARD THEORY]**: a standard result from gravitational-potential theory, spherical geometry, linear elasticity, or continuum mechanics. It was not invented for this project, but it is not being attributed to one of the project papers.
- **[DERIVED HERE]**: an algebraic result obtained in this guide/notebook by combining source equations with standard identities. It is not an equation quoted from a paper.
- **[MODELING CHOICE]**: an assumption, normalization, parameter choice, or transfer of a literature coefficient introduced specifically for this SAFOD AWD exercise.

Most important provenance statement: Model A and Model B are *not* models published by Niu et al., Métivier & Conrad, De Fazio et al., or Hole et al. They are constructions made for this SAFOD AWD analysis using published theory, published measurements, standard mechanics, and explicit modeling assumptions.

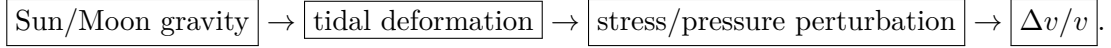
De Fazio et al. (1973) supplies historical and physical context for tidal velocity measurements and source repeatability, but it does not supply a governing equation used to generate Model A or Model B. Takano et al. (2014) is discussed for context but is not used to generate either model in the current notebook.

1 The point of the whole exercise

The easiest way to understand this project is to forget the detailed equations for a moment and think of the Earth as a physical system with an input and an output.



For the tidal problem,



The central scientific question is therefore not merely “can I see the Moon in my data?” The useful question is:

If I know the amplitude and timing of a small external forcing, how strongly does my seismic observable respond to it?

That response is a *sensitivity* or transfer coefficient. If the input is stress, one may write

$$S_{\sigma} \equiv \frac{\Delta v/v}{\Delta \sigma}, \quad (1)$$

with units of Pa^{-1} . If the input is pressure, one may instead define

$$C_P \equiv \frac{\Delta v/v}{\Delta P}. \quad (2)$$

These coefficients are valuable because they tell us how sensitive a particular rock volume, fracture network, borehole mode, or monitoring configuration is to mechanical perturbation.

The key idea: geophysicists care about tides because tides provide a naturally imposed, repeatable, predictable, very small mechanical perturbation. If a seismic quantity responds coherently to that known input, the response can reveal the mechanical sensitivity of the system. The tide is the probe; the crustal response is the thing of interest.

This is exactly the logic used in early active-source tidal velocity studies such as De Fazio et al. (1973), in the SAFOD experiment of Niu et al. (2008), and in later ambient-noise work such as Takano et al. (2014).

2 Why tidal signals are useful in geophysics

2.1 A naturally repeating calibration signal

The gravitational forcing from the Moon and Sun is deterministic to very high accuracy. It occurs at known frequencies and phases, including long-period, diurnal, and semidiurnal bands. This makes it

fundamentally different from an earthquake or an uncontrolled environmental transient.

A small signal that repeats coherently can be extracted statistically even when any one cycle is noisy. Conceptually this is similar to lock-in detection: if the expected forcing has a known phase, one can ask whether the observation contains a component at that same phase.

2.2 The forcing is small enough to probe the near-background state

Large laboratory stress changes can push a rock far from its ambient state. Tidal stresses are tiny by comparison. That makes them useful for measuring local sensitivity around the current in-situ stress state.

This matters because seismic velocity is not perfectly independent of stress. Cracks and compliant pores can open or close under small changes in effective stress, changing the elastic moduli and therefore the wave speed. Niu et al. (2008) emphasize that velocity stress sensitivity depends on crack density and confining pressure, while Takano et al. (2014) interpret their tidal velocity response in terms of opening and closure of cracks or pores.

2.3 Tides can probe more than seismic velocity

Across geophysics, tidal forcing is used as a small known input to study quantities such as strain, gravity, fluid pressure, permeability, poroelastic response, tremor triggering, and seismic velocity. The common logic is

$$\text{known periodic forcing} \rightarrow \text{measured response} \rightarrow \text{inferred sensitivity/physics.} \quad (3)$$

For the present AWD project, the most relevant use is much narrower: tides provide a physically meaningful small-signal benchmark against which to compare the empirically measured $\Delta v/v$ sensitivity of the repeated-source experiment.

3 The quantities that must not be mixed up

Several different physical quantities appear in this problem. Most confusion comes from treating them as interchangeable when they are not.

Quantity	Typical symbol	Meaning
Displacement	$\mathbf{u}$	How far a material point moves. Units: m.
Strain	ϵ	Fractional deformation or displacement gradient. Dimensionless.
Stress	σ	Internal force per unit area. Units: Pa.
Pressure	P or p	Isotropic normal loading; in porous media fluid pressure also affects effective stress. Units: Pa.
Velocity change	$\Delta v/v$	Fractional change in a seismic velocity or apparent-mode velocity. Dimensionless.

Sensitivity	S_σ, C_P	Ratio between a measured fractional velocity change and an independently known forcing.
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A strain of 10^{-8} does *not* mean a velocity change of 10^{-8} . To go from strain to velocity change one needs additional physics.

4 Why stress can change seismic velocity

For a homogeneous isotropic elastic material,

$$V_P = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}, \quad V_S = \sqrt{\frac{\mu}{\rho}}, \quad (4)$$

where K is bulk modulus, μ is shear modulus, and ρ is density.

For a small perturbation,

$$\frac{\Delta V_P}{V_P} \approx \frac{1}{2} \frac{\Delta(K + \frac{4}{3}\mu)}{K + \frac{4}{3}\mu} - \frac{1}{2} \frac{\Delta\rho}{\rho}, \quad (5)$$

and

$$\frac{\Delta V_S}{V_S} \approx \frac{1}{2} \frac{\Delta\mu}{\mu} - \frac{1}{2} \frac{\Delta\rho}{\rho}. \quad (6)$$

Stress can therefore affect velocity if it changes the effective elastic moduli. In cracked rock this can happen because compression closes compliant cracks, increasing stiffness, while unloading can open them and decrease stiffness.

For the 2026 AWD experiment, however, the measured quantities are *apparent velocities of coherent modes*. In general one should think of

$$v_{\text{app}} = F(K, \mu, \rho, P_{\text{fluid}}, \text{fractures, borehole geometry, coupling, } \omega, \dots), \quad (7)$$

rather than assuming $v_{\text{app}} = V_P$ or V_S . A first-order perturbation can be written schematically as

$$\frac{\Delta v_{\text{app}}}{v_{\text{app}}} \approx \sum_j \frac{\partial \ln v_{\text{app}}}{\partial q_j} \Delta q_j, \quad (8)$$

where the q_j are whatever physical properties actually control that mode.

This is why a coefficient measured for one wave type cannot automatically be transferred to another as a universal rock property.

5 Where the solid-Earth tide comes from

5.1 Gravity from an external body

Let an external body (Moon or Sun) have mass M_b and geocentric distance D_b . A point inside or on Earth has position vector $\mathbf{r}$ with $r < D_b$. The gravitational potential of the body is

$$U_b(\mathbf{r}) = \frac{GM_b}{|\mathbf{D}_b - \mathbf{r}|}. \quad (9)$$

For $r/D_b < 1$, this can be expanded in Legendre polynomials:

$$U_b = \frac{GM_b}{D_b} \sum_{n=0}^{\infty} \left(\frac{r}{D_b} \right)^n P_n(\cos \theta_b), \quad (10)$$

where θ_b is the angle between the local position vector and the direction to the external body.

The first terms have different meanings:

- $n = 0$: a spatially constant potential. A constant potential produces no deformation.
- $n = 1$: a nearly uniform acceleration of Earth as a whole toward the external body. In an Earth-centered frame this rigid-body acceleration is removed.
- $n = 2$: the first term that describes *differential* gravity across Earth and therefore the leading deformation-producing tide.

This is why the degree-2 term is the natural starting point.

5.2 The degree-2 tide-generating potential

[STANDARD THEORY] The equations in this subsection are the classical $n = 2$ term of the Legendre expansion of the gravitational potential of an external point mass. They are not equations copied from Métivier and Conrad (2008); that paper provides the spherical-harmonic/Love-number framework and the justification for focusing on degree 2.

At Earth's surface $r = R_E$, the leading tide-generating potential from body b is

$$W_b(t) = \frac{GM_b}{D_b(t)^3} R_E^2 P_2[\cos \theta_b(t)], \quad (11)$$

with

$$P_2(x) = \frac{1}{2} (3x^2 - 1). \quad (12)$$

The total first-order luni-solar forcing used in the notebook is

$$W(t) = W_{\text{Moon}}(t) + W_{\text{Sun}}(t). \quad (13)$$

5.3 What “degree 2” and “order m ” mean

In a spherical-harmonic representation, degree n controls the broad spatial complexity on the sphere, while order m describes longitudinal variation. For degree 2, the familiar tidal families are

$$\begin{aligned} m = 0 & : \text{long-period,} \\ |m| = 1 & : \text{diurnal,} \\ |m| = 2 & : \text{semidiurnal.} \end{aligned} \quad (14)$$

[**SOURCE – Métivier & Conrad (2008)**] Métivier and Conrad state that these degree-2 long-period, diurnal, and semidiurnal terms together account for approximately 99.98% of the total luni-solar tide-generating-potential *power*. This is the justification for using degree 2 as a first-order forcing model. It is not a claim that degree 2 reproduces 99.98% of the local fault stress at SAFOD.

6 How Earth responds: Love numbers

Earth is not rigid. The tidal potential produces radial and horizontal displacements. [**SOURCE – M’etivier & Conrad (2008), eq. (3)**] In the classical Love-number description used by that paper, for a degree- n potential component,

$$\mathbf{u} = \frac{W_n}{g} [h_n Y_n \mathbf{e}_r + l_n \nabla_1 Y_n], \quad (15)$$

where h_n describes radial displacement and l_n (the Shida number) describes horizontal displacement. The notation ∇_1 denotes the dimensionless surface angular gradient.

[**SOURCE – M’etivier & Conrad (2008), eq. (6)**] For degree 2, the quasi-static PREM values used in the notebook are

$$h_2 = 0.603741, \quad l_2 = 0.084010, \quad k_2 = 0.298231, \quad d_2 = 1.156394. \quad (16)$$

The notebook’s deformation calculations use h_2 and l_2 . The values are quasi-static: inertial effects are neglected, and small frequency dependence due to rotation and anelasticity is not included.

7 Deriving the scalar areal-strain proxy

The first notebook model reduces the degree-2 deformation to a scalar horizontal areal strain.

[**STANDARD THEORY**] For a small patch of the spherical surface, the horizontal areal strain can be written schematically as

$$\varepsilon_A = \nabla_s \cdot \mathbf{u}_h + \frac{2u_r}{R_E}, \quad (17)$$

where $\mathbf{u}_h$ is horizontal displacement and u_r is radial displacement.

[SOURCE + STANDARD IDENTITY] Using the Love-number displacement relation from M'etivier and Conrad together with the standard spherical-harmonic eigenvalue identity,

$$u_r = \frac{h_2}{g} W, \quad (18)$$

and the horizontal divergence of the degree-2 displacement is related to the spherical Laplacian. Spherical harmonics satisfy

$$\nabla_1^2 Y_{nm} = -n(n+1)Y_{nm}. \quad (19)$$

For $n = 2$,

$$n(n+1) = 2(3) = 6. \quad (20)$$

[DERIVED HERE] Therefore the degree-2 horizontal divergence contributes a factor $-6l_2$, while the radial expansion contributes $2h_2$. The scalar proxy below is a derivation used in this guide/notebook; it is not an equation quoted directly from M'etivier and Conrad:

$$\boxed{\varepsilon_{A,\text{proxy}}(t) = (2h_2 - 6l_2) \frac{W(t)}{gR_E}}. \quad (21)$$

For the adopted PREM Love numbers,

$$2h_2 - 6l_2 = 2(0.603741) - 6(0.084010) = 0.703422. \quad (22)$$

During the 2026 AWD window, the notebook gives a peak-to-peak scalar areal-strain proxy of approximately

$$\Delta\varepsilon_{A,\text{pp}} \approx 7.87 \times 10^{-8}, \quad (23)$$

or about 79 nanostrain.

Important: Equation (21) is a deformation model. It is not yet a stress model and not yet a $\Delta v/v$ model.

8 Model A: a SAFOD-calibrated first-order $\Delta v/v$ template

8.1 Niu et al.’s empirical SAFOD stress sensitivity

[**SOURCE – Niu et al. (2008)**] Niu et al. observed that their SAFOD travel-time changes tracked barometric pressure. After correcting for temperature, they reported a velocity perturbation of about 3×10^{-4} associated with a barometric pressure change of about 1.3 kPa and used a linear regression to obtain

$$S_\sigma \approx 2.4 \times 10^{-7} \text{ Pa}^{-1}. \quad (24)$$

[**SOURCE – Niu et al. (2008)**] They also calculated that solid-Earth tidal stress during their observing interval varied within approximately

$$\Delta\sigma_{\text{tide}} \sim 240 \text{ Pa}. \quad (25)$$

[**DERIVED HERE FROM TWO NIU VALUES**] Multiplying the two reported Niu values gives the characteristic SAFOD scale used in this guide

$$\left| \frac{\Delta v}{v} \right|_{\text{Niu scale}} \sim S_\sigma \Delta\sigma_{\text{tide}} \quad (26)$$

$$= (2.4 \times 10^{-7} \text{ Pa}^{-1}) (240 \text{ Pa}) \quad (27)$$

$$= 5.76 \times 10^{-5} \quad (28)$$

$$= 0.00576\%. \quad (29)$$

Niu et al. concluded that the corresponding tidal travel-time perturbation would be close to their measurement error and therefore predicted it to be undetectable in their experiment.

8.2 Turning the scale into a June 2026 time series

[**MODELING CHOICE**] The 240 Pa number was not calculated for June 2026. To obtain a first-order time history, the notebook takes the modeled June 2026 degree-2 areal-strain *shape*, removes its mean, and normalizes it. This normalization is introduced here; it is not from Niu et al.:

$$\tau(t) = \frac{\varepsilon_{A,\text{proxy}}(t) - \overline{\varepsilon_{A,\text{proxy}}}}{\max_t |\varepsilon_{A,\text{proxy}}(t) - \overline{\varepsilon_{A,\text{proxy}}}|}. \quad (30)$$

Thus $|\tau(t)| \leq 1$. [**MODELING CHOICE**] The stress template is defined by imposing Niu’s 240 Pa characteristic scale onto that normalized June 2026 shape:

$$\Delta\sigma_A(t) = 240 \tau(t) \text{ Pa}, \quad (31)$$

[**MODELING ASSUMPTION**] The velocity template then assumes that Niu’s barometric stress-sensitivity coefficient can be applied linearly to this imposed stress template:

$$\boxed{\left(\frac{\Delta v}{v}\right)_A(t) = S_\sigma \Delta\sigma_A(t)}. \quad (32)$$

The maximum magnitude is therefore fixed to

$$\max_t \left| \left(\frac{\Delta v}{v}\right)_A \right| = 5.76 \times 10^{-5} = 0.00576\%. \quad (33)$$

8.3 Exactly what Model A means

Model A combines two different pieces of information:

1. **Timing and relative shape:** the June 2026 degree-2 forcing calculation.
2. **Amplitude:** Niu et al.’s SAFOD stress scale and empirical stress sensitivity.

It is therefore best understood as a *SAFOD-calibrated order-of-magnitude timing template constructed in this work*, not as a model published by Niu et al. It is not an independently calculated June 2026 stress tensor. The sign should also not be overinterpreted because the scalar strain template does not prove which stress component controls the 2026 apparent modes.

9 Model B: tensor-aware elastic scenario

[**MODEL CONSTRUCTED HERE**] Model B as a whole is a construction made for this SAFOD AWD analysis; it is not a model taken from any one paper. It asks a different question:

Can we retain the directional tidal deformation, convert it to an explicit elastic stress tensor, project that stress onto a plausible San Andreas Fault plane, and then estimate $\Delta v/v$?

This introduces more physics, but also more assumptions.

9.1 Local north-east displacement

[**SOURCE – M’etivier & Conrad (2008), eq. (3), rewritten into local components here**] Let ϕ be latitude and λ longitude. The degree-2 Love-number displacement components used in the notebook are

$$u_r = \frac{h_2}{g} W, \quad (34)$$

$$u_N = \frac{l_2}{g} \frac{\partial W}{\partial \phi}, \quad (35)$$

and

$$u_E = \frac{l_2}{g \cos \phi} \frac{\partial W}{\partial \lambda}. \quad (36)$$

The derivatives are angular derivatives; division by R_E enters when converting displacement gradients to physical strain.

9.2 Horizontal strain tensor on a spherical surface

The local horizontal strain tensor is

$$\boldsymbol{\varepsilon}_h = \begin{bmatrix} \varepsilon_{NN} & \varepsilon_{NE} \\ \varepsilon_{NE} & \varepsilon_{EE} \end{bmatrix}. \quad (37)$$

[STANDARD THEORY] Using north-east coordinates on a sphere, the small-strain kinematic formulas implemented in the notebook are

$$\varepsilon_{NN} = \frac{1}{R_E} \left(\frac{\partial u_N}{\partial \phi} + u_r \right), \quad (38)$$

$$\varepsilon_{EE} = \frac{1}{R_E} \left[\frac{1}{\cos \phi} \frac{\partial u_E}{\partial \lambda} + u_r - u_N \tan \phi \right], \quad (39)$$

and

$$\varepsilon_{NE} = \frac{1}{2R_E} \left[\frac{\partial u_E}{\partial \phi} + u_E \tan \phi + \frac{1}{\cos \phi} \frac{\partial u_N}{\partial \lambda} \right]. \quad (40)$$

[DERIVED-HERE CONSISTENCY CHECK] A crucial consistency check is

$$\varepsilon_{NN} + \varepsilon_{EE} \approx \varepsilon_{A,\text{proxy}}. \quad (41)$$

The notebook's numerical mismatch is only about 2.9×10^{-8} *relative*, confirming that the tensor trace reproduces the scalar areal-strain formula to numerical precision.

9.3 Strain to stress: isotropic plane stress

[STANDARD THEORY + MODELING CHOICE] The equations below are standard isotropic linear-elastic plane-stress Hooke-law relations. The equations themselves are standard; the

choice to apply a surface plane-stress approximation to this SAFOD tide calculation is ours. The horizontal stresses are

$$\sigma_{NN} = \frac{E}{1 - \nu^2} (\varepsilon_{NN} + \nu \varepsilon_{EE}), \quad (42)$$

$$\sigma_{EE} = \frac{E}{1 - \nu^2} (\varepsilon_{EE} + \nu \varepsilon_{NN}), \quad (43)$$

and

$$\sigma_{NE} = \frac{E}{1 + \nu} \varepsilon_{NE}. \quad (44)$$

[MODELING CHOICE] The baseline scenario uses

$$E = 50 \text{ GPa}, \quad \nu = 0.25, \quad (45)$$

and also evaluates $E = 20\text{--}70$ GPa while holding the other assumptions fixed.

The phrase *scenario envelope* is important: this is not a statistical confidence interval.

9.4 Why plane stress is only a surface approximation

[STANDARD THEORY; APPLIED HERE AS A MODELING APPROXIMATION]

Plane stress imposes

$$\sigma_{DD} = \sigma_{ND} = \sigma_{ED} = 0, \quad (46)$$

where D is the local downward direction. This is motivated by the traction-free boundary at Earth's surface. It is not an exact description at hundreds of meters or kilometers depth. Therefore the resulting stress tensor should not be described as the exact tidal stress at the SAFOD fiber or active shear zones.

9.5 Projecting the stress tensor onto a fault

[STANDARD THEORY] Let $\mathbf{n}$ be a unit vector normal to the fault plane. The traction acting on the plane is

$$\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}. \quad (47)$$

The normal stress is

$$\boxed{\sigma_n = \mathbf{n}^T \boldsymbol{\sigma} \mathbf{n}} \quad (48)$$

and the shear traction vector is

$$\boldsymbol{\tau} = \boldsymbol{t} - \sigma_n \boldsymbol{n}. \quad (49)$$

[**SOURCE – Hole et al. (2001), simplified into a geometry scenario here**] The notebook uses a geometry scenario with strike 140° , dip 70° , and southwest dip direction. In a north-east-down coordinate system, a strike unit vector can be written

$$\boldsymbol{s} = \begin{bmatrix} \cos \alpha \\ \sin \alpha \\ 0 \end{bmatrix}, \quad (50)$$

and a down-dip vector

$$\boldsymbol{d} = \begin{bmatrix} \cos \delta \cos \beta \\ \cos \delta \sin \beta \\ \sin \delta \end{bmatrix}, \quad (51)$$

where α is strike, δ is dip, and β is dip-direction azimuth. The normal is then

$$\boldsymbol{n} = \frac{\boldsymbol{s} \times \boldsymbol{d}}{\|\boldsymbol{s} \times \boldsymbol{d}\|}. \quad (52)$$

[**MODELING CHOICE**] The code uses a tension-positive elastic stress convention and then converts the normal component to compression-positive form:

$$\sigma_{n,\text{comp}} = -\sigma_n. \quad (53)$$

[**MODELING ASSUMPTION USING A SOURCE-DERIVED COEFFICIENT**] After removing the 24-hour mean, Model B applies Niu’s coefficient. The coefficient itself is from Niu et al.; applying it to the fault-normal stress calculated here is our assumption:

$$\boxed{\left(\frac{\Delta v}{v}\right)_B(t) = S_\sigma \Delta \sigma_{n,\text{comp}}(t)}. \quad (54)$$

Niu et al. inferred S_σ from barometric loading of their own SAFOD seismic observable. They did not establish that the same coefficient applies to fault-normal tidal stress, nor to the 2026 Nano or Deep apparent modes. That transfer is therefore a modeling assumption, not a literature result.

For the $E = 50$ GPa baseline, the notebook gives

$$\max |\Delta \sigma_{n,\text{comp}}| \approx 1516 \text{ Pa}, \quad (55)$$

and

$$\max \left| \left(\frac{\Delta v}{v} \right)_B \right| \approx 3.64 \times 10^{-4} = 0.0364\%. \quad (56)$$

For the $E = 20\text{--}70$ GPa scenario range,

$$\max \left| \left(\frac{\Delta v}{v} \right)_B \right| \approx 1.46 \times 10^{-4} \text{ to } 5.09 \times 10^{-4}, \quad (57)$$

or approximately 0.0146%–0.0509%.

9.6 Why Model B is not automatically better

Model B is more explicit, but every added step introduces assumptions:

$$\boxed{W \rightarrow \varepsilon \rightarrow \sigma \rightarrow \sigma_n \rightarrow \Delta v/v}. \quad (58)$$

The astronomical part, W , is comparatively well constrained. The largest uncertainties are the local mechanical and constitutive arrows to the right.

Specifically, Model B assumes that

- a surface degree-2 strain calculation is adequate for the relevant depth;
- a homogeneous isotropic elastic modulus is meaningful for the shallow SAFOD setting;
- ocean loading and other loading corrections can be neglected for this first pass;
- the chosen fault plane is the mechanically relevant orientation;
- the relevant velocity responds to fault-normal stress;
- Niu’s barometric velocity-stress coefficient transfers to the 2026 apparent mode.

The last assumption is especially important. A mathematically computed σ_n is not useful unless σ_n is actually the physical variable controlling the measured mode.

10 What the AWD experiment actually calibrated

The injection-recovery analysis does not calibrate stress sensitivity. It calibrates the *measurement sensitivity to an imposed fractional apparent-velocity change*.

For the finalized 15–30 Hz Deep analysis,

$$\begin{array}{ll} \text{Deep outbound:} & 95\% \text{ null threshold} = 1.84 \times 10^{-3}, \text{ reliable level} = 5 \times 10^{-3}, \\ \text{Deep return:} & 95\% \text{ null threshold} = 3.05 \times 10^{-3}, \text{ reliable level} = 1 \times 10^{-2}, \\ \text{Nano:} & 95\% \text{ null threshold} = 3.25 \times 10^{-3}, \text{ reliable level} = 1 \times 10^{-2}. \end{array} \quad (59)$$

The reliable-detection criterion is the smallest tested $|\epsilon|$ for which at least 95% of trials both exceed the empirical null threshold and recover the correct sign.

Thus the best demonstrated level is

$$\boxed{|\Delta v/v|_{\text{reliable}} = 5 \times 10^{-3} = 0.5\%} \quad (60)$$

for the Deep outbound branch.

Compared with Model A,

$$\frac{5 \times 10^{-3}}{5.76 \times 10^{-5}} \approx 86.8. \quad (61)$$

Therefore the Niu-scale tidal signal is about 87 times smaller than the best demonstrated reliable-detection level.

11 An “equivalent stress” intuition—with a major caveat

If, purely for intuition, one pretends that the Niu coefficient applies directly to the 2026 Deep apparent mode, then

$$\Delta\sigma_{\text{equiv}} = \frac{|\Delta v/v|}{S_\sigma}. \quad (62)$$

For the Deep outbound reliable level,

$$\Delta\sigma_{\text{equiv}} = \frac{5 \times 10^{-3}}{2.4 \times 10^{-7}} \quad (63)$$

$$\approx 2.08 \times 10^4 \text{ Pa} \quad (64)$$

$$\approx 20.8 \text{ kPa}. \quad (65)$$

For a 1% reliable level,

$$\Delta\sigma_{\text{equiv}} \approx 41.7 \text{ kPa}. \quad (66)$$

For the Deep outbound empirical null threshold 1.84×10^{-3} ,

$$\Delta\sigma_{\text{equiv,null}} \approx 7.7 \text{ kPa}. \quad (67)$$

These are **not** measured stress-detection thresholds. They are only a unit conversion using Niu’s coefficient. They become physically meaningful only if the 2026 mode is shown to have the same stress sensitivity.

12 How to calculate your own coefficient

This is the most direct route from the present work toward a physically interpretable monitoring experiment.

12.1 The simplest pressure coefficient

Suppose each AWD burst gives an apparent-velocity estimate

$$y_i = \left(\frac{\Delta v}{v} \right)_i, \quad (68)$$

and an independent sensor gives a pressure perturbation

$$x_i = \Delta P_i. \quad (69)$$

A simple linear model is

$$y_i = \beta_0 + C_P x_i + \eta_i, \quad (70)$$

where C_P is the empirical pressure sensitivity and η_i contains noise and unmodeled effects.

With unweighted least squares, the fitted slope is

$$\boxed{\hat{C}_P = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}}. \quad (71)$$

Its units are Pa^{-1} .

12.2 Including drift

For a short field experiment, both pressure and $\Delta v/v$ may contain slow trends. A more useful model is

$$y_i = \beta_0 + \beta_1 t_i + C_P \Delta P_i + \eta_i. \quad (72)$$

In matrix form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta}, \quad (73)$$

with

$$\mathbf{X} = \begin{bmatrix} 1 & t_1 & \Delta P_1 \\ 1 & t_2 & \Delta P_2 \\ \vdots & \vdots & \vdots \\ 1 & t_N & \Delta P_N \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ C_P \end{bmatrix}. \quad (74)$$

With weights $\mathbf{W}$, the weighted least-squares estimate is

$$\boxed{\hat{\boldsymbol{\beta}} = \left(\mathbf{X}^T \mathbf{W} \mathbf{X} \right)^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}}. \quad (75)$$

The third component of $\hat{\boldsymbol{\beta}}$ is the estimated pressure coefficient.

12.3 Different coefficients answer different physical questions

One could estimate

$$C_{\text{baro}} = \frac{\Delta v/v}{\Delta P_{\text{atm}}}, \quad (76)$$

$$C_{\text{bh}} = \frac{\Delta v/v}{\Delta P_{\text{borehole}}}, \quad (77)$$

or

$$C_{\text{WHP}} = \frac{\Delta v/v}{\Delta P_{\text{wellhead}}}. \quad (78)$$

These are empirical transfer coefficients. They do not require pretending that pressure equals fault stress.

A true stress coefficient

$$C_{\sigma} = \frac{\Delta v/v}{\Delta \sigma} \quad (79)$$

requires a defensible independent estimate of the relevant stress perturbation.

12.4 Effective stress if pore pressure is the mechanism

If the physical hypothesis is that pore pressure changes effective normal stress, a common poroelastic form is

$$\boldsymbol{\sigma}' = \boldsymbol{\sigma} - \alpha_B p \mathbf{I}, \quad (80)$$

for a compression-positive convention, where α_B is Biot's coefficient. The normal effective-stress perturbation would be

$$\Delta\sigma'_n = \Delta\sigma_n - \alpha_B\Delta p. \quad (81)$$

A corresponding empirical sensitivity could be written

$$C_{\sigma'} = \frac{\Delta v/v}{\Delta\sigma'_n}. \quad (82)$$

But this requires knowing or modeling $\Delta\sigma_n$, Δp , and α_B . Wellhead pressure is not automatically pore pressure at the sensing location.

13 From an empirical coefficient to a physical detection threshold

Once a coefficient is measured for the actual AWD mode, the $\Delta v/v$ sensitivity can be converted into a forcing sensitivity.

If

$$\frac{\Delta v}{v} = C_P\Delta P, \quad (83)$$

then the pressure perturbation needed to reach a chosen velocity threshold is

$$\boxed{\Delta P_{\text{detect}} \approx \frac{|\Delta v/v|_{\text{detect}}}{|C_P|}}. \quad (84)$$

For example, if a future controlled experiment measured

$$|C_P| = 1 \times 10^{-7} \text{ Pa}^{-1}, \quad (85)$$

then the Deep outbound reliable level 5×10^{-3} would correspond to

$$\Delta P_{\text{detect}} \approx \frac{5 \times 10^{-3}}{10^{-7}} = 5 \times 10^4 \text{ Pa} = 50 \text{ kPa}. \quad (86)$$

This example is hypothetical; the point is the logic.

14 Directly fitting a tidal template to the AWD $\Delta v/v$ series

There is a second question distinct from predicting the expected amplitude:

Does the measured $\Delta v/v$ time series contain a component shaped like the tide?

A simple regression is

$$y_i = A \tau(t_i) + b_0 + b_1 t_i + \eta_i, \quad (87)$$

where

- y_i is the measured per-burst apparent-mode $\Delta v/v$,
- $\tau(t_i)$ is a normalized tidal predictor,
- A is the fitted tidal amplitude,
- $b_0 + b_1 t_i$ represents an offset and linear drift.

The 2026 Deep analysis used this form with a 999-realization surrogate null. No significant tidal signal was found. On all 46 available Deep bursts, the best paired estimate gave a 95% upper limit of about

$$1.03 \times 10^{-3}. \quad (88)$$

The important lesson is that the amplitude uncertainty was larger than one would predict from per-burst scatter alone. Over only about 22 hours, a diurnal tidal shape is not well separated from a slow linear drift. This is a design limitation, not evidence that the tide physically does not exist.

15 Why a 24-hour experiment is awkward for tides

Suppose the signal contains a diurnal term

$$s(t) = A \cos(\omega t + \phi), \quad \omega \approx \frac{2\pi}{24 \text{ h}}. \quad (89)$$

Over a short fraction of one cycle, a cosine can resemble a constant plus a linear trend. A Taylor expansion illustrates this:

$$\cos(\omega t + \phi) \approx \cos \phi - \omega t \sin \phi - \frac{1}{2} \omega^2 t^2 \cos \phi + \dots. \quad (90)$$

If the record is only about one day long, the first two terms can be difficult to distinguish from an unknown offset and instrumental drift. Longer records provide repeated cycles and therefore much greater leverage for separating tide from drift.

This is why “better per-burst precision” and “better tidal detectability” are not the same thing.

16 How the two calibration problems fit together

There are two separate calibrations in this project.

16.1 Calibration 1: measurement sensitivity

The injection-recovery experiment answers

$$\boxed{\text{How small a change in apparent velocity can the analysis recover?}} \quad (91)$$

For Deep outbound, the answer is a reliably recovered tested level of 0.5%.

16.2 Calibration 2: physical sensitivity

A controlled pressure or stress experiment would answer

$$\boxed{\text{How much does this apparent velocity change for a known physical perturbation?}} \quad (92)$$

That is the coefficient C_P or C_σ .

Only after both are known can one convert an instrumental/analysis threshold into a physically meaningful pressure or stress threshold:

$$\boxed{\text{physical forcing} \xrightarrow{C_P \text{ or } C_\sigma} \Delta v/v \xrightarrow{\text{AWD sensitivity}} \text{detect / do not detect.}} \quad (93)$$

17 What a controlled pressure experiment would ideally look like

A clean future experiment would record, simultaneously,

1. repeated AWD bursts;
2. borehole, formation, or injection pressure at sufficiently high precision;
3. barometric pressure;
4. temperature;
5. any operational changes that could alter coupling or borehole conditions.

For each burst i , estimate

$$y_i = (\Delta v/v)_i. \quad (94)$$

Then fit a model such as

$$y_i = \beta_0 + \beta_1 t_i + C_{\text{bh}} \Delta P_{\text{bh},i} + C_{\text{atm}} \Delta P_{\text{atm},i} + C_T \Delta T_i + \eta_i. \quad (95)$$

The goal is not to include every possible predictor automatically; the purpose of writing the general model is to make confounding explicit.

If pressure changes propagate to the sensing region with a lag, an exploratory model could include

$$y(t) \approx C_P \Delta P(t - \tau_P) + \dots, \quad (96)$$

where τ_P is a hydraulic response time. A more physical model would replace the simple lag with a pressure-diffusion or poroelastic transfer function.

18 The hierarchy of tidal models

It helps to think of tidal modeling as a ladder rather than as one “correct” equation.

- Level 1: Tide-generating potential only.** Compute the astronomical forcing $W(t)$.
- Level 2: Surface deformation.** Use Love numbers to compute displacement or strain.
- Level 3: Surface elastic stress scenario.** Convert strain to stress with an assumed elastic model.
- Level 4: Depth-specific stress.** Use depth-dependent Earth response and local structure.
- Level 5: Loading corrections.** Add ocean loading, atmospheric loading, hydrology as needed.
- Level 6: Path/fault projection.** Resolve the tensor onto the mechanically relevant direction.
- Level 7: Constitutive seismic response.** Relate the relevant stress/pressure perturbation to the specific measured seismic mode.

The current notebook contains Levels 1–3 and an illustrative Level 6 projection, plus an empirical coefficient borrowed from Niu for Level 7. It does not contain a complete Levels 4–7 site-specific model.

19 What is robust and what is not

19.1 Relatively robust

- The Sun and Moon produce a degree-2-dominated solid-Earth tide.
- Degree-2 long-period, diurnal, and semidiurnal terms account for about 99.98% of tide-generating-potential power.

- The adopted quasi-static PREM Love numbers are standard first-order values.
- The June 2026 scalar strain proxy is of order tens of nanostrain.
- Niu et al. reported $S_\sigma \approx 2.4 \times 10^{-7} \text{ Pa}^{-1}$ and a tidal stress scale of about 240 Pa at SAFOD during their experiment.
- The 2026 AWD injection-recovery calibration gives a best reliable tested apparent-velocity change of 0.5% on Deep outbound.

19.2 Model-dependent

- the exact June 2026 stress tensor at borehole depth;
- the stress component that controls the Deep guided mode;
- the sign and amplitude of the true tidal $\Delta v/v$ response of that mode;
- whether Niu’s barometric coefficient transfers to the 2026 modes;
- the conversion of the AWD $\Delta v/v$ threshold into a pressure/stress threshold.

20 A compact group-meeting explanation

A concise verbal explanation can be built from four statements:

1. “Earth tides are useful because they are a known, periodic, very small mechanical forcing. If seismic velocity responds to them, the response tells us something about the stress sensitivity of the medium or waveguide.”
2. “I first model the degree-2 Sun-Moon forcing and convert it to a surface strain using PREM Love numbers. That gives me the timing and deformation scale.”
3. “To get $\Delta v/v$, I either calibrate the amplitude with Niu’s SAFOD stress sensitivity (Model A), or I explicitly convert the directional strain tensor to an elastic stress scenario and project it onto a fault plane before applying the same coefficient (Model B).”
4. “Both calculations predict changes below the AWD experiment’s demonstrated 0.5% reliable level. More importantly, the injection-recovery analysis tells me what the experiment can measure; a future controlled pressure perturbation could give me my own pressure-to- $\Delta v/v$ coefficient and turn that measurement threshold into a physical pressure threshold.”

21 Equation map

The entire chain can be summarized as follows.

Astronomical forcing

$$W_b = \frac{GM_b}{D_b^3} R_E^2 P_2(\cos \theta_b), \quad P_2(x) = \frac{3x^2 - 1}{2}. \quad (97)$$

Surface deformation

$$\varepsilon_{A,\text{proxy}} = (2h_2 - 6l_2) \frac{W_{\text{Sun}} + W_{\text{Moon}}}{gR_E}. \quad (98)$$

Model A

$$\tau(t) = \frac{\varepsilon_{A,\text{proxy}} - \overline{\varepsilon_{A,\text{proxy}}}}{\max |\varepsilon_{A,\text{proxy}} - \overline{\varepsilon_{A,\text{proxy}}}|}, \quad (99)$$

$$\Delta\sigma_A = 240\tau \text{ Pa}, \quad (100)$$

$$(\Delta v/v)_A = S_\sigma \Delta\sigma_A. \quad (101)$$

Model B

$$W \rightarrow \{u_r, u_N, u_E\} \rightarrow \{\varepsilon_{NN}, \varepsilon_{EE}, \varepsilon_{NE}\} \rightarrow \{\sigma_{NN}, \sigma_{EE}, \sigma_{NE}\}, \quad (102)$$

$$\sigma_n = \mathbf{n}^T \boldsymbol{\sigma} \mathbf{n}, \quad (103)$$

$$(\Delta v/v)_B = S_\sigma \Delta\sigma_{n,\text{comp}}. \quad (104)$$

Your own future coefficient

$$(\Delta v/v)_i = \beta_0 + \beta_1 t_i + C_P \Delta P_i + \eta_i, \quad (105)$$

$$\Delta P_{\text{detect}} \approx \frac{|\Delta v/v|_{\text{detect}}}{|C_P|}. \quad (106)$$

22 Questions to ask yourself when looking at any tidal- $\Delta v/v$ paper

1. What is the measured seismic observable: direct P , S , coda, surface wave, tube/Stoney wave, guided mode, or an apparent moveout?
2. What is the forcing variable: strain, stress, atmospheric pressure, pore pressure, or a modeled tidal component?

3. Is the reported coefficient empirical or theoretical?
4. Is it local to the site and wave type, or transferred from another experiment?
5. What sign convention is used for compression and strain?
6. Is the tide model surface-only or depth-specific?
7. Are ocean and atmospheric loading included?
8. Is the observed record long enough to distinguish diurnal/semidiurnal forcing from drift?
9. Is the result a detection, an upper limit, or only an order-of-magnitude benchmark?

Those questions are more important than memorizing any individual coefficient.

23 References

References

- [1] De Fazio, T. L., Aki, K., & Alba, J. (1973). Solid Earth Tide and Observed Change in the In Situ Seismic Velocity. *Journal of Geophysical Research*, 78, 1319–1322.
- [2] Métivier, L., & Conrad, C. P. (2008). Body tides of a convecting, laterally heterogeneous, and aspherical Earth. *Journal of Geophysical Research: Solid Earth*, 113, B11405.
- [3] Niu, F., Silver, P. G., Daley, T. M., Cheng, X., & Majer, E. L. (2008). Preseismic velocity changes observed from active source monitoring at the Parkfield SAFOD drill site. *Nature*, 454, 204–208.
- [4] Takano, T., Nishimura, T., Nakahara, H., Ohta, Y., & Tanaka, S. (2014). Seismic velocity changes caused by the Earth tide: Ambient noise correlation analyses of small-array data. *Geophysical Research Letters*, 41, 6131–6136.